

## 列主元高斯消去法实验报告



代码：

```
#include <stdio.h>
#include <math.h>
#define n 3

int main() {
    double a[n+1][n+1] = {{0,0,0,0},
                           {0,0.2641,0.1735,0.8642},
                           {0,0.9411,-0.0175,0.1463},
                           {0,-0.8641,-0.4243,0.0711}};

    double b[n+1] = {0,-0.7521,0.6310,0.2501};
    double l[n+1][n+1], x[n+1], sum;
    int k, i, j, p, ll;

    for (k = 1; k <= n-1; k++) {
        p = k;
        for (i = k+1; i <= n; i++) {
            if (fabs(a[i][k]) > fabs(a[p][k]))
                p = i;
        }
        // 交换第 p 行和第 k 行
        for (j = 1; j <= n; j++) {
            double temp = a[p][j];
            a[p][j] = a[k][j];
            a[k][j] = temp;
        }
        double temp = b[p];
        b[p] = b[k];
        b[k] = temp;

        // 进行消元
        for (i = k+1; i <= n; i++) {
            l[i][k] = a[i][k] / a[k][k];
            b[i] = b[i] - l[i][k] * b[k];
            for (j = k+1; j <= n; j++) {
                a[i][j] = a[i][j] - l[i][k] * a[k][j];
            }
        }
    }
}
```

```

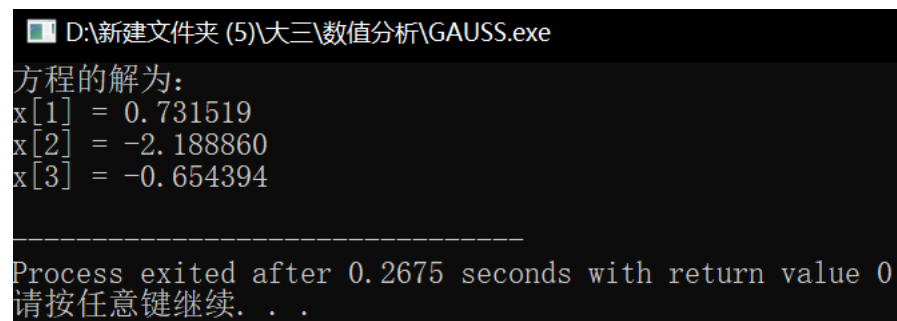
// 回代计算解
x[n] = b[n] / a[n][n];
for (k = n-1; k >= 1; k--) {
    sum = 0;
    for (l1 = k+1; l1 <= n; l1++) {
        sum = sum + a[k][l1] * x[l1];
    }
    x[k] = (b[k] - sum) / a[k][k];
}

printf("方程的解为: \n");
for (k = 1; k <= n; k++) {
    printf("x[%d] = %f\n", k, x[k]);
}

return 0;
}

```

运行结果:



```

D:\新建文件夹 (5)\大三\数值分析\GAUSS.exe
方程的解为:
x[1] = 0.731519
x[2] = -2.188860
x[3] = -0.654394

-----
Process exited after 0.2675 seconds with return value 0
请按任意键继续. . .

```